

Adversarial Cooperation

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Abstract—Adversarial Cooperation seeks solutions to problems in game theory using cryptographic protocols without a trusted referee, and hardware for their execution. The aim is cooperation that preserves private information and the strategic advantages it supports. We seek resources to develop, analyze, implement, and demonstrate these protocols and their hardware.

Index Terms—Applied cryptography, cooperation, zero knowledge, multiparty computation, hardware.

I. THE COOPERATION PROBLEM

The central question is: *how can we construct and demonstrate cryptographic protocols that let adversarial parties cooperate without a referee, while retaining their strategic advantage?* A party should be able to take part in a common game or verify an agreed result without handing its cards, strategy, or other secrets to an opponent or a trusted intermediary. We find private and public institutions are not aware of how exposed they are.

The privacy objective is to reveal nothing about private inputs beyond the agreed outputs and public information. Each protocol must specify its adversary model, assumptions, and privacy and correctness requirements.

Candidate tools include cryptographic hashes and transcript hashing; message authentication codes, authenticated encryption, and digital signatures; commitments [1], coin flipping, and secret sharing; oblivious transfer, zero-knowledge proofs [2], and multiparty computation [3]; verifiable shuffles, threshold cryptography, private set operations, and verifiable computation; randomness extraction, key agreement, and password-authenticated key exchange. Selection depends on the problem's requirements and assumptions. Prior work on replacing mediators in strategic games also informs this research [4]. No new primitive or general solution is claimed; each proposed construction requires its own analysis.

II. GAMES AND COOPERATION PROBLEMS

The following cases define initial research targets:

Rock–Paper–Scissors

Fix each player's choice before either learns the other's move, then verify the completed round's outcome.

Tic-Tac-Toe

Demonstrate a private strategy avoids defeat under specified game conditions, without disclosing move-selection rules.

Poker

Investigate dealing and winner verification without a trusted dealer, without revealing private cards beyond what the agreed outcome implies.

Stag Hunt

Coordinate mutually beneficial action when each party's willingness to cooperate depends on the other's participation.

Joint Optimization

Optimize an agreed objective using private inputs or constraints, revealing only the agreed result.

Each case requires its own specification and analysis. Whether more complex cooperation can be composed from simpler, elemental games and protocols remains an open question.

III. FROM SPECIFICATION TO HARDWARE

The goal is to take formally defined protocols through inspectable executable representations into specialized hardware that participants can independently inspect and reproduce. Candidate methods include C reference code, Boolean circuits, register-transfer-level (RTL) descriptions in Verilog or VHDL, and FPGA evaluation; choices depend on the protocol.

Translation correctness and correspondence between specification, code, circuits, and devices are research tasks. Prior work on verified high-level synthesis provides a foundation [5]. Evaluation would address functional behavior, costs, fabrication and device integrity, and physical leakage or faults. Functional correspondence alone does not establish protection against such physical attacks.

IV. RESOURCES AND SUPPORT

We seek funding, research collaborators, computing resources, FPGA boards, and measurement equipment. Support would enable dedicated research, independent cryptographic review, implementation, and hardware experiments.

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